

LABORATORY ACTIVITY

PatutiM: Grid Survival Game

Sprite Animation and Gameplay Enhancement

Laboratory Overview

The supplied PatutiM project is a grid-based survival/path game. The player moves Patuti one cell at a time through a hidden valid path. Correct cells are revealed as the player explores; an incorrect cell costs one life. The game ends when the player reaches the goal or loses all lives. The starter version uses one idle image. In this laboratory, you will first trace the existing game loop, then enhance it with the supplied movement sprites.

Learning Outcomes

- Load multiple PNG sprites as embedded resources and manage them as animation frames.
- Implement frame-based animation for idle, left, right, jump/up, and dock/down movement.
- Synchronize sprite animation with grid movement without breaking path validation.
- Preserve boundary checking, life management, and game-state transitions.
- Test the game systematically and document defects and corrections.

Sprite Asset Preview

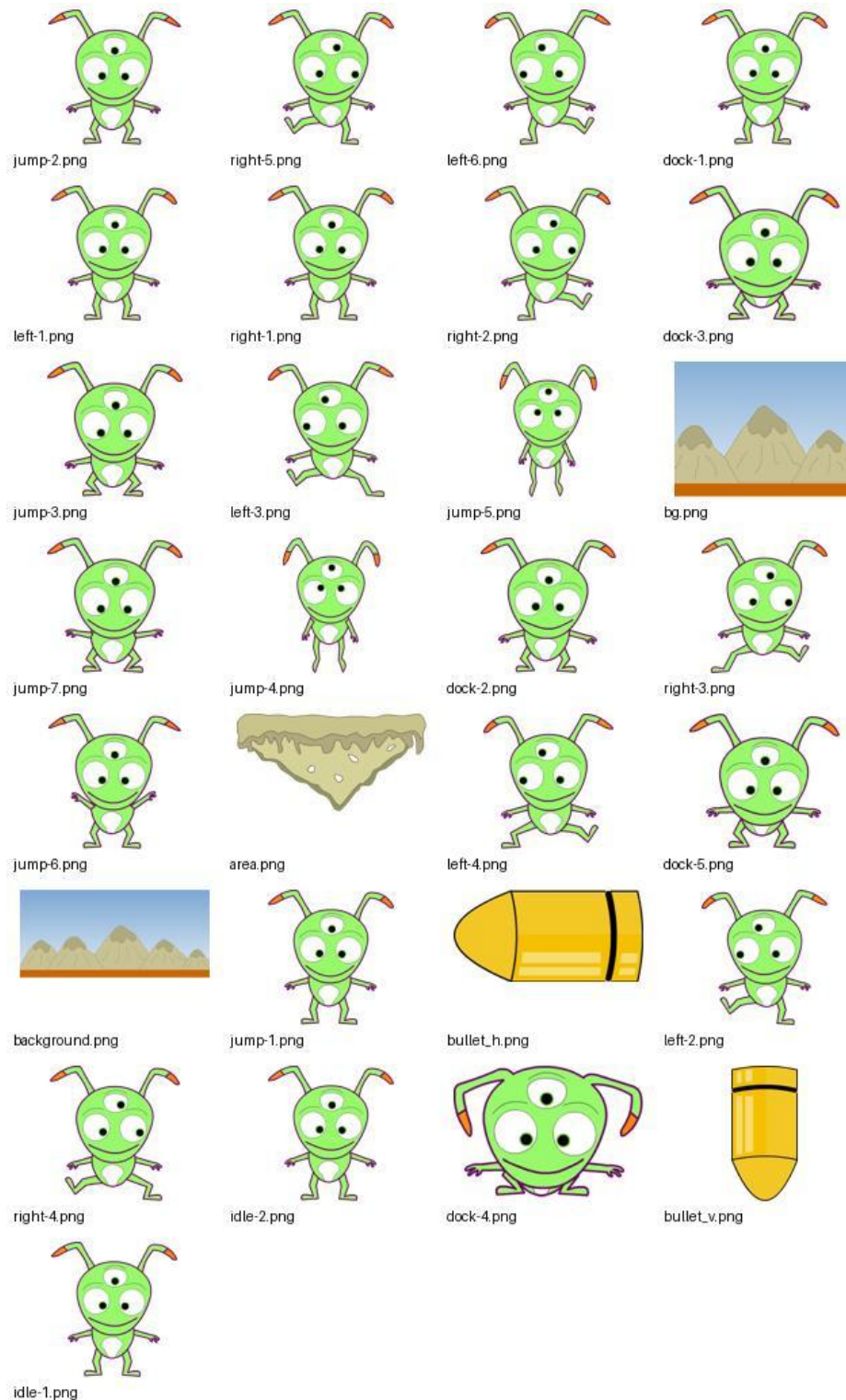


Figure 1. Supplied sprites from moves.zip.

Required work uses the character frames: idle, left, right, jump, and dock. Background, area, and bullet assets are reserved for the extension challenge.

Game Flow

1. The four buttons send command parameters 1–4 for Up, Down, Left, and Right.
2. A valid button press changes a target coordinate by one grid-cell width.
3. During each repaint, Patuti moves a few pixels toward the target.
4. When the target is reached, movement stops and the new cell is checked against mat[.,].
5. A true matrix cell belongs to the safe path; a false cell removes one life.
6. The path/allpath stacks control which boxes remain visible and therefore reveal explored cells.
7. Reaching the target area wins; reaching zero lives ends the game.

Existing Rules

- One accepted button press moves exactly one grid cell.
- Input is ignored while Patuti is moving.
- Left/right boundary protection still works.
- A wrong cell removes exactly one life.
- Life icons match the lives variable.
- Zero lives disables controls and shows game over.
- Reaching the goal disables controls and shows Success.
- Hidden-path behavior still works after animation is added.

Required: Reset / Play Again

Add a Reset or Play Again button that restores the game without restarting the application.

- lives returns to 5;
- Patuti returns to the start;
- path and allpath are cleared;
- movement/validation flags are reset;
- playerState becomes Idle;
- movement buttons are enabled;
- prompt text is reset.

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